Tech Quest Treasure Hunt: Design Document

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Overview

The treasure hunt is a 6-minute dungeon crawl where a team of 5 plays one knight descending into the Cavern of Code, answering quick questions to beat enemies, defeating a final boss, and opening a chest that holds the next clue.

- **Event:** Tech Quest, run by the BCA department for pre-university (PU) students.
- **Goal:** show students, from any stream, that computer science is approachable and fun, so they consider BCA.
- **Audience:** teams of 5 PU students (about 300 students, so about 60 teams); no prior knowledge needed, and commerce students should feel at home.
- **Format:** one projector screen, no keyboards for players, one control head (you) running the session.
- **Theme:** a Skyrim-style ruin. One continuous corridor of small enemies, then a boss, then the reward.
- **Winning:** the fastest total times reach the top of the leaderboard; the top 10 teams advance and the top 3 win prizes.

Gameplay flow

The knight moves forward through one unbroken corridor; every enemy is a single question, and clearing all of them leads to the boss and the chest.

1. **Start:** the control head enters the team name and presses Start. The timer begins when the first enemy appears.
2. **Enemies:** each enemy appears as a sprite with one question. The team discusses, and one person calls the answer aloud. The control head taps correct or wrong.
3. **Correct answer:** the enemy is defeated, a letter or digit drops, and the knight advances.
4. **Wrong answer:** the knight takes a hit, 10 seconds are added to the team's time, and the next enemy appears after a short pause, so nobody is ever stuck.
5. **Boss, The Summarizer:** a mixed recap of all four domains. The team picks easy, medium, or hard questions to wear down the boss's 100 HP (see Hit points and damage).
6. **Chest:** defeating the boss opens the chest and reveals the next clue in the treasure hunt. The timer stops and the team's time goes on the leaderboard.

Enemies are asked in the same order for every team: easiest first, hardest last, as in a real dungeon.

Hit points and damage

Every question carries a difficulty tag that sets how hard the knight hits. These are starting values to tune in the dry run.

Difficulty	Damage	Used for
Easy	20 HP	Opening enemies, warm-up questions
Medium	30 HP	Mid-run enemies, boss round
Hard	50 HP	Late-zone enemies, boss round

- In the four zones, one correct answer still defeats the enemy. Damage sets the size of the knight's hit: a small slash for easy, a big punch for hard.
- The boss has 100 HP and needs 2 to 5 correct answers to fall.
- For each boss question, the team chooses the difficulty. Hard ends the fight faster, but a wrong answer still adds 10 seconds.

Content plan

The run has 12 enemies in four zones plus a boss fight, about 6 minutes in total at 15 to 20 seconds per enemy. Final counts depend on the dry run.

Zone	Domain	Enemies	Example questions	Time
1. Crypt of Cipher	Data: cipher, storage, types	3	Decode a shift-cipher word; binary to a letter using a wall chart; pick the data type for a price, a name, a yes/no	~1 min
2. Forge of Logic	Programming	3	What does this 3-line pseudocode print; put shuffled steps in order; spot the bug	~1 min
3. Gate of Shields	Cybersecurity	3	Spot the scam among three UPI or parcel messages; strongest of four passwords; what to do when someone asks for an OTP	~1 min
4. Heart of the Cave	AI	3	Spot AI imagery (which image is AI-made); prompting challenge (pick the prompt that gets the described result); recognise AI activity (which everyday tools use AI)	~1 min
Boss: The Summarizer	All four	2 to 5	Mixed recap across all four domains; the team picks easy, medium, or hard, and the boss's 100 HP wears down	~1 min

Rules for writing questions

- Every question is answerable by logic or common sense; no prior CS knowledge is needed.
- Each answer is a single word, letter, or choice, so the control head can mark it in one tap.
- Difficulty rises inside each zone and across the run.
- Every enemy has 2 or 3 variants of equal difficulty with different content (a different cipher word, a different scam message). The app picks one at random per team.
- The first zone must also work as a cold start, since all teams begin there.

Scoring and leaderboard

A team's score is its total time from the first enemy to the chest opening, plus penalties; the lowest time ranks first.

Event	Effect on time
Wrong answer	+10 seconds
Enemy skipped by the control head	+20 seconds
Tie on total time	Fewer wrong answers wins

- The leaderboard updates live and shows rank, team name, and total time.
- The top 10 teams are highlighted, since they advance.
- With about 60 teams, the top 10 on the leaderboard advance and the top 3 teams win prizes.

App and screens

The app is a browser page on the projector for players and a separate control panel for the control head; players never touch a keyboard.

Screen	Who sees it	What it shows
Dungeon view	Players, on the projector	Cave background, pixel-art knight, current enemy sprite, question, timer, three hearts
Control panel	Control head only	Start, correct, wrong, skip enemy, reset run, add time, team name entry
Leaderboard	Players, between runs or in a corner	Live rank, team name, total time, top teams highlighted
Victory screen	Players	Chest opening, next clue, final time

Look and feel

- Dark cave with torchlight glow; each zone has its own accent colour.
- One shared timer and one "enemy defeated" animation across all zones, so it feels like a single descent.
- Large text readable from the back of the room.

Technical notes

- Questions and variants live in one data file, so they can be edited without touching the app.
- The control panel can run on a phone or laptop on the same network as the projector page.
- Results are saved locally as well as shown, so a lost connection never loses the leaderboard.

Fairness, risks and open questions

The main risk is that later teams overhear answers from earlier ones, which would make the leaderboard unreliable.

Risk	Mitigation
Teams overhear answers	2 or 3 variants per enemy, picked at random; hold waiting teams away from the room
A team stalls on one enemy	Control head skip (+20 seconds)
Run exceeds 6 minutes	Cut to 8 or 9 enemies if the dry run is slow
Image questions need preparation	Source real and AI-made image pairs ahead of time for Spot AI Imagery
Projector or network failure	Keep a printed backup question set and a manual stopwatch

Open questions

- What happens to the top 10 teams next: a final round, or do the top 3 on the leaderboard simply take the prizes?
- About 60 teams at 6 minutes each is about 6 hours on one screen; how many stations or screens will run at once?
- Confirm the damage design: easy 20, medium 30, hard 50, and a 100 HP boss.
- AI zone: who prepares the image pairs and the prompting challenge options?

Next steps

- [ ] Settle the open questions above
- [ ] Write all questions, variants, and answer keys for the four zones and the boss
- [ ] Choose the clue the chest reveals
- [ ] Source or create the AI and real image pairs
- [ ] Build the dungeon view, control panel, and leaderboard
- [ ] Dry run with a small group and time it
- [ ] Trim or add enemies so the run fits 6 minutes